

### Bioengineered robotic hand that will actually feel

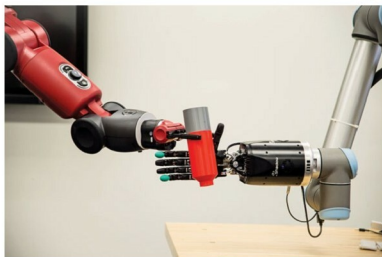
Scientists from Florida Atlantic University (FAU) and the University of Utah School of Medicine are developing a bioengineered robotic hand that will grow and adapt to its environment. This 'living' robot will have its own peripheral nervous system directly linking robotic sensors and actuators. The team is creating a living pathway from the robot's touch sensation to the user's brain to help amputees control the robotic hand. A neuroprosthesis platform

will enable them to explore how neurons and behaviour can work together to regenerate the sensation of touch in an artificial limb.

At the core of this project is a cutting-edge robotic hand and arm developed in FAU's College of Engineering and Computer Science. Just like human fingertips, the robotic hand is equipped with numerous sensory receptors that respond to changes in the environment. Controlled by a human, it can

sense pressure changes, interpret the information it is receiving and interact with various objects. It adjusts its grip based on an object's weight or fragility. But the real challenge is figuring out how to send that information back to the brain using living residual neural pathways to replace those that have been damaged or destroyed by trauma.

"When the peripheral nerve is cut or damaged, it uses the rich electrical activity that tactile receptors create to restore itself. We want to examine how the fingertip sensors can help damaged or severed nerves regenerate," said Erik Engeberg, principal investigator. "To accomplish this, we are going to directly connect these living nerves in vitro and then electrically stimulate them on a daily basis with sensors from the robotic hand to see how the nerves grow and regenerate while the hand is operated by limb-absent people."



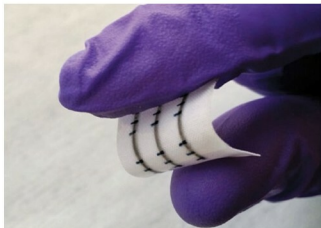
*Just like human fingertips, the robotic hand is equipped with numerous sensory receptors that respond to changes in the environment (Image credit: Florida Atlantic University)*

### Fully integrated circuits printed directly onto fabric

Researchers have demonstrated how graphene, a two-dimensional form of carbon, can be directly printed onto fabric to produce integrated electronic circuits that are comfortable to wear and can survive up to 20 cycles in a typical washing machine.

The new textile electronic devices are based on low-cost, sustainable and scalable inkjet printing of inks based on graphene and other two-dimensional materials, and are produced by standard processing techniques. The results by researchers from the University of Cambridge, with colleagues in Italy and China, are published in the journal *Nature Communications*.

Based on earlier work on the formulation of graphene inks for printed electronics, the team designed low-boiling point inks, which were directly printed onto polyester fabric. Additionally, they found that modifying the roughness of the



*A sample circuit printed on fabric (Image credit: Felice Torrisi, University of Cambridge)*